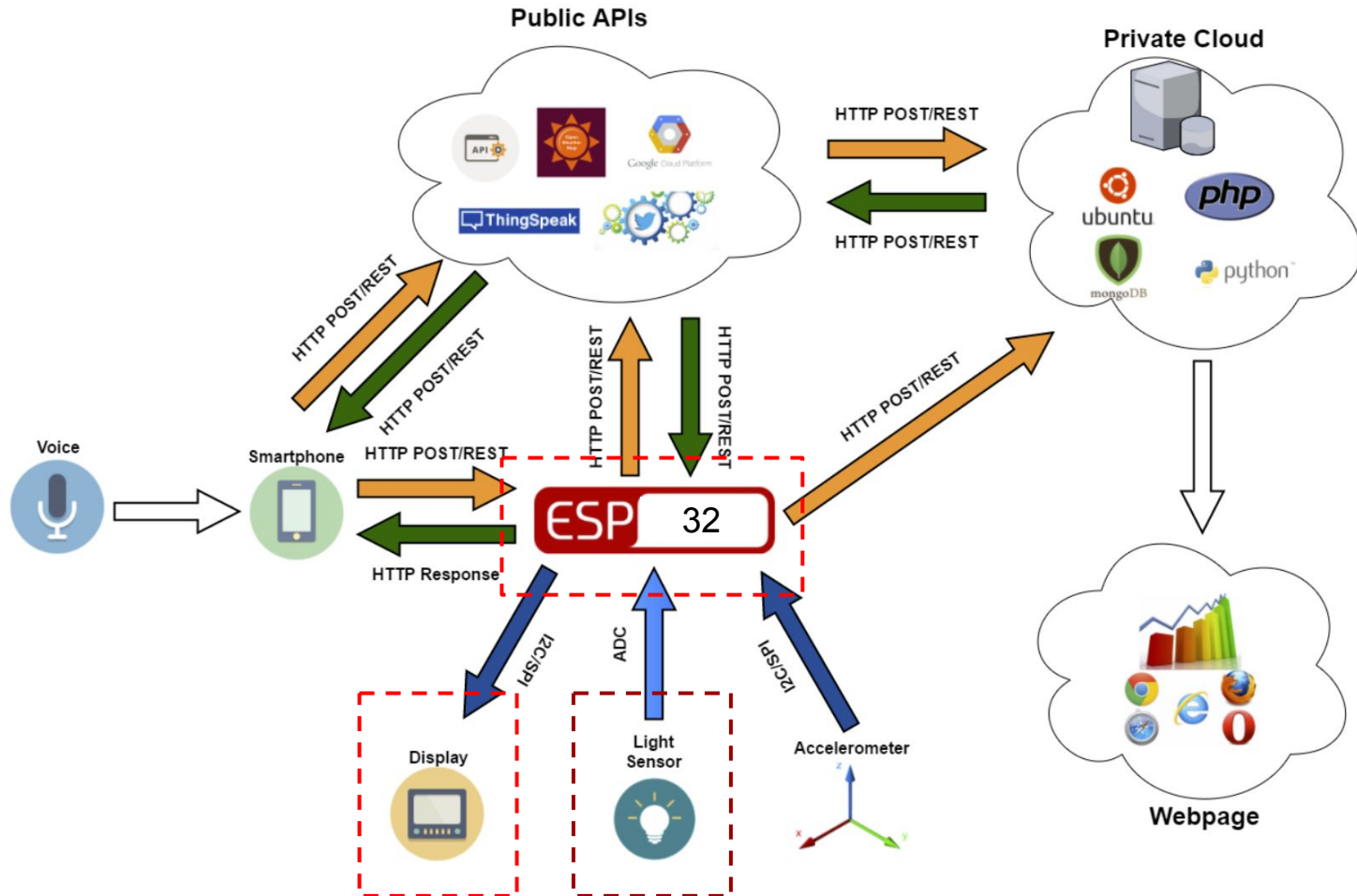
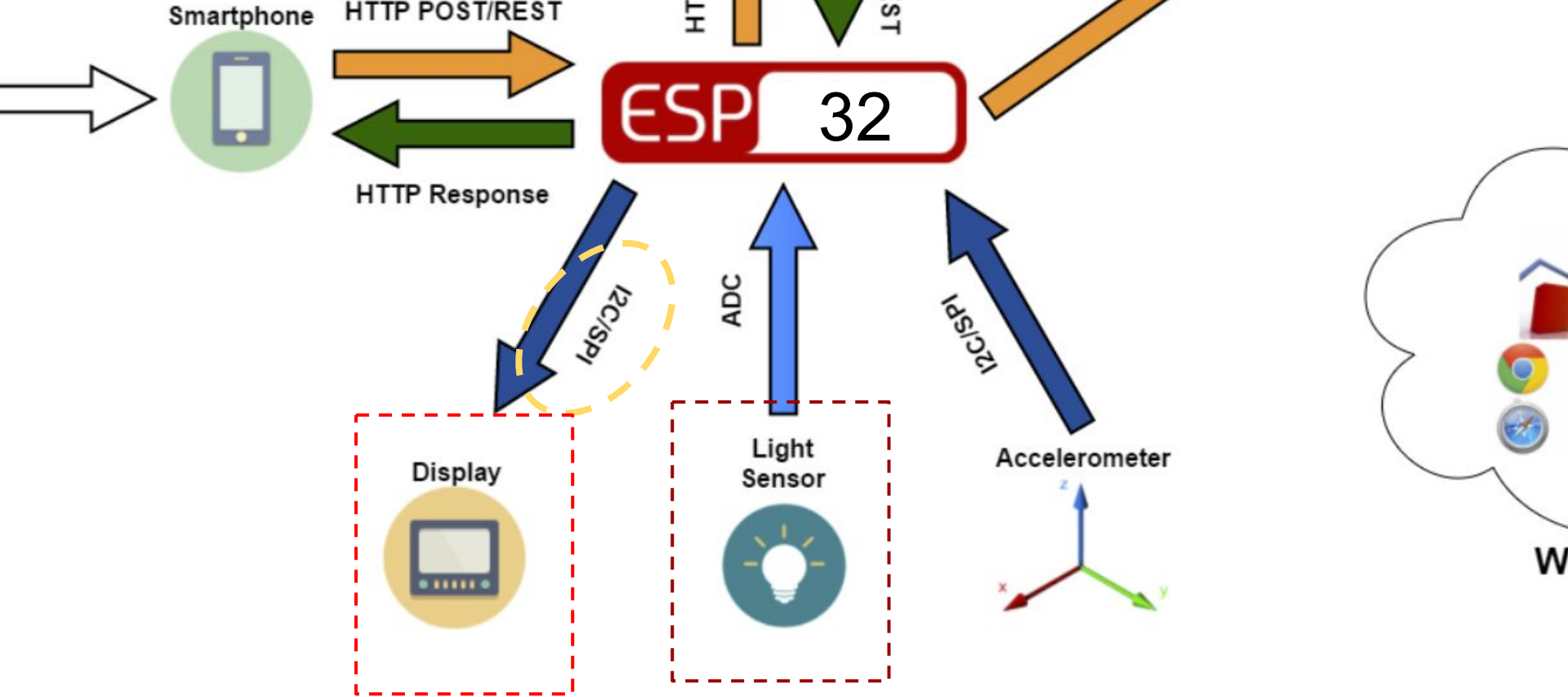


AIoT - Intelligent and Connected Systems

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Lab 3





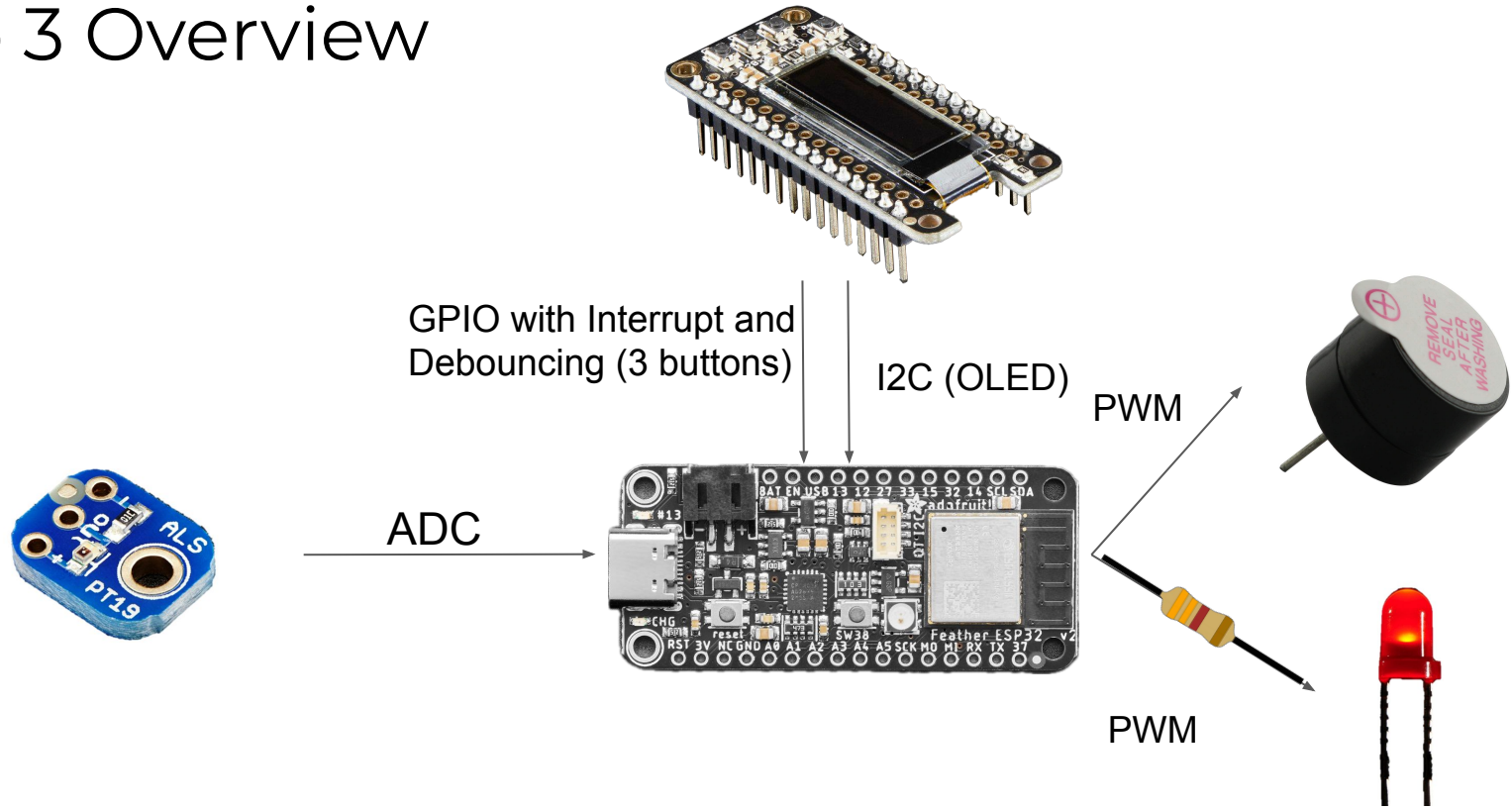
Lab3: Apply I2C protocol to transmit data to/from display

Requirements

1. Your watch can **update time** properly like a normal watch
2. Your watch can **setup time**, and it keeps working properly
3. Your watch can **setup the alarm**, and the **alarm works** when it is the time
4. You correctly **debounce all the buttons**



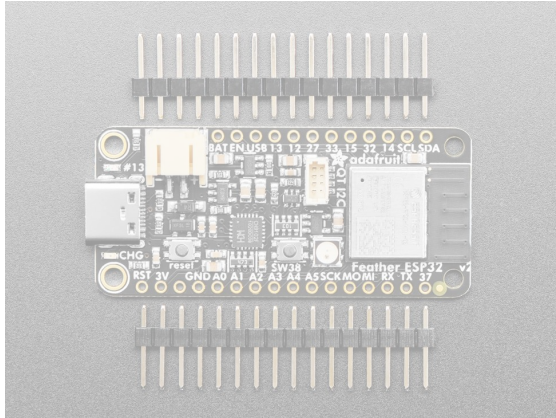
Lab 3 Overview



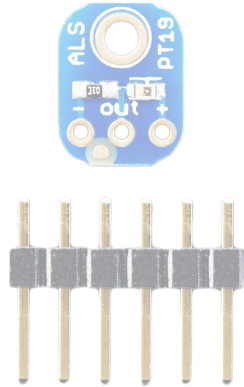
- Pins and GPIO: <https://docs.micropython.org/en/latest/esp32/quickref.html#pins-and-gpio>
- ESP32 Pinouts: <https://learn.adafruit.com/adafruit-esp32-feather-v2/pinouts>
- OLED Pinouts: <https://learn.adafruit.com/adafruit-oled-featherwing/pinouts>
- OLED Library: <https://docs.micropython.org/en/latest/esp8266/tutorial/ssd1306.html> (SSD1306)

Soldering

Lab 2 (last week)

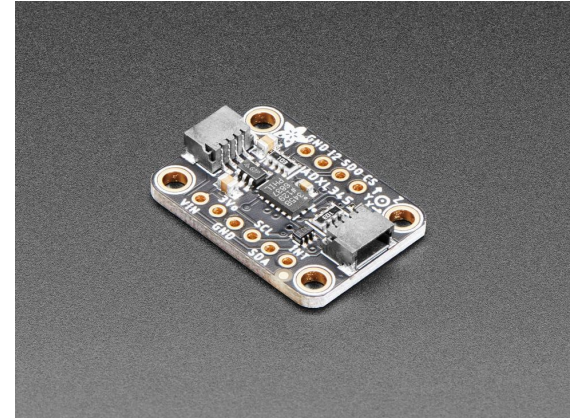


Feather HUZZAH



Light Sensor

Lab 4 (next week)

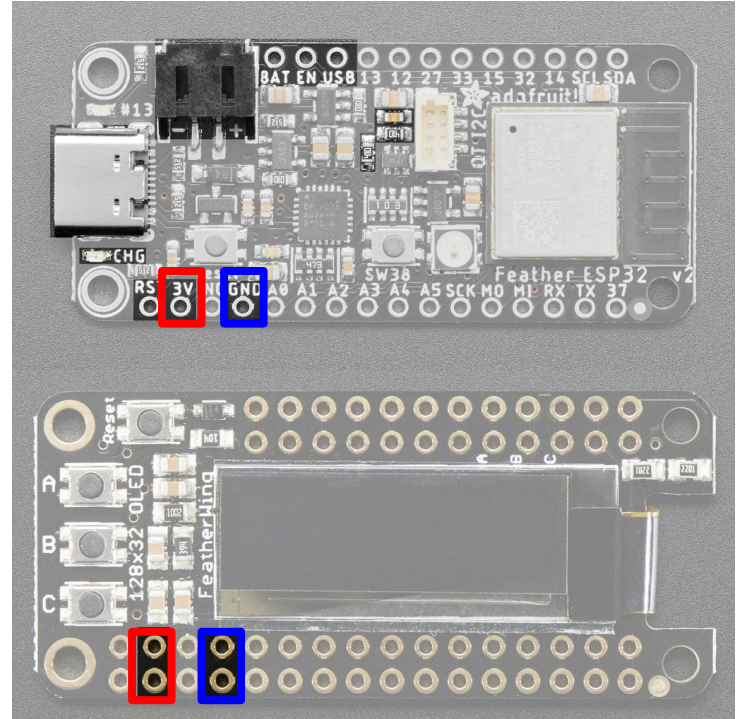


Accelerometer

Powering OLED

OLED displays do not have a backlight, and are fairly low power, this display will draw about 10mA when in use.

The display uses 3V power and logic so we just connect to the 3V and GND pins from the feather, as indicated right.

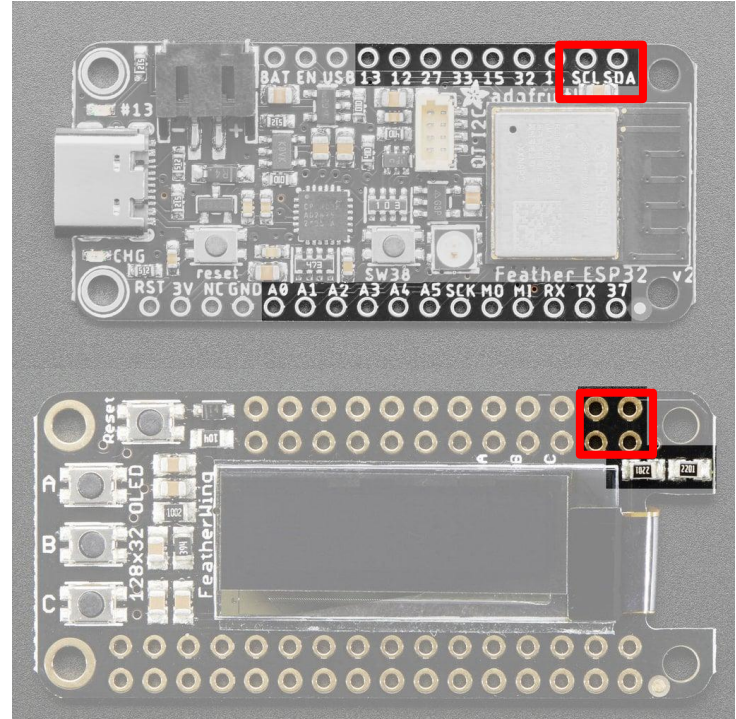


Connecting OLED

The Huzzah and OLED pins are corresponded one by one.

SCL - This is the I2C clock pin.
It is also pin **20**.

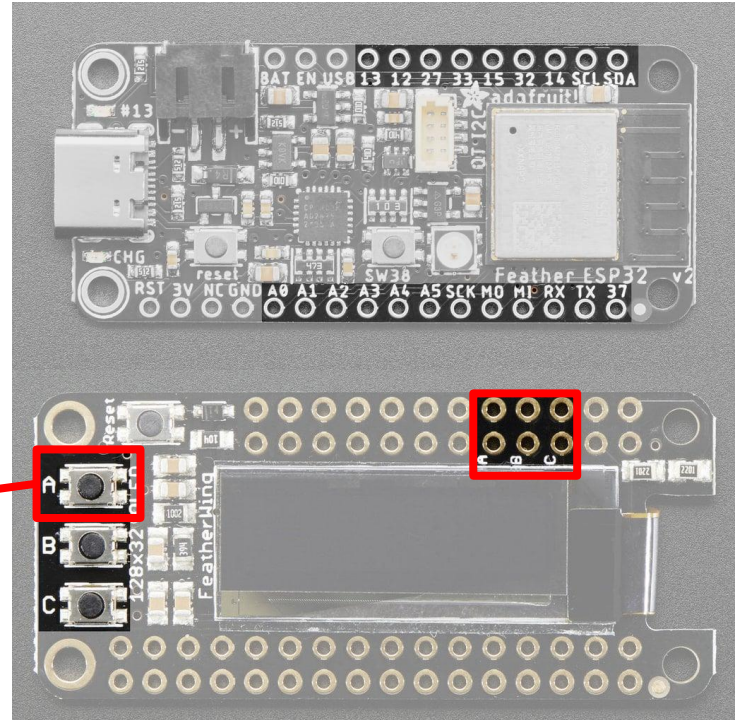
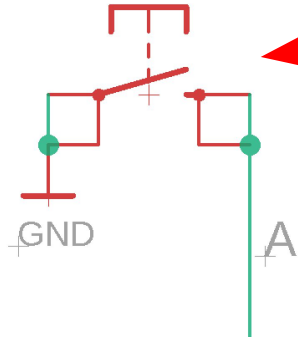
SDA - This is the I2C data pin.
It is also pin **22**.



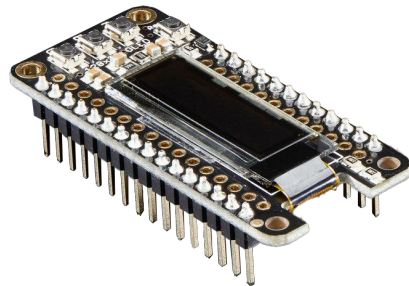
Connecting OLED

Buttons A, B, C are connected to GND on the OLED PCB.

Use any GPIO on HUZZAH that has **interrupt** and **internal pull-up** capability.



OLED Example



Installing the SSD1306 Library in MicroPython

The **SSD1306** library is **not included** in the standard MicroPython firmware.

To use the OLED display, you should first download the provided `ssd1306.py` file from Coursework.

Once downloaded, upload it to your ESP32 using **mpfshell**:

“put [ssd1306.py](#)”

(If you know about it, you can also use the package installer provided by Arduino)

```
from machine import Pin, I2C
import ssd1306

# using default address 0x3C
i2c = I2C(sda=Pin(22), scl=Pin(20))
display = ssd1306.SSD1306_I2C(128, 32, i2c)

# Print Hello World on the first line:
display.text('Hello, World!', 0, 0, 1)
display.show()
```

- OLED Library: <https://docs.micropython.org/en/latest/esp8266/tutorial/ssd1306.html> (SSD1306)

Lab 3 Checkpoints

1. Display the time on the OLED display; the time should update like a regular watch. Be able to change time through the OLED display buttons.
2. Adjust screen brightness based on ambient light.
3. Add alarm functionality; Be able to set the alarm through OLED display and when the alarm goes off, a visual and audio notification should go off.
4. Combine 1, 2, and 3.